

Abstract: Synoptic Context and Surface Evolution of the North Atlantic Cyclone : Satellite/Radar Imagery and Surface Analyses; Contextualizing Met.3D Visualizations

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Understanding how major storms develop requires looking beyond 2D maps to explore the atmosphere as a complex 3D structure. This study provides the synoptic foundation for the NAWDIC campaign by analyzing the North Atlantic environment from 5 to 11 February 2026. Rather than viewing a storm as a static event, this analysis tracks the transition from broad baroclinic instability into mature extratropical systems, providing the framework needed to identify high-potential targets for research flights.

The period begins on 5 February with a strong temperature gradient across the North Atlantic, setting the stage for cyclogenesis. By midday on 7 February, a powerful low-pressure system is visible over the open ocean. Using DWD and UKMO standards, we identify an intense pressure gradient west of the European coast that generates surface winds reaching $35\text{--}50\text{ m s}^{-1}$. These winds act as a conveyor, pulling a dense moisture plume from the subtropics that manifests as distinct, heavy precipitation bands.

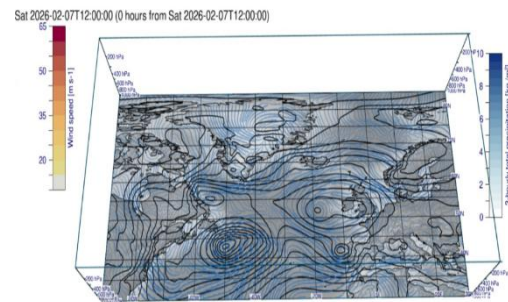


Figure 1: Integrated Met.3D visualization showing 3-hourly total precipitation (shaded) and wind speed isosurfaces at multiple pressure levels. Arrows indicate the direction of the low-level jet and moisture transport.

Sat 2026-02-07T12:00:00 (0 hours from Sat 2026-02-07T12:00:00)

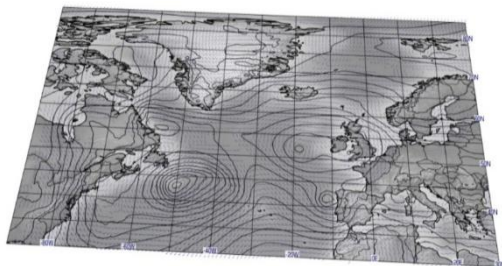


Figure 2: Synoptic-scale surface pressure analysis and cloud cover for the North Atlantic sector. The visualization shows the horizontal extent of the cyclone and its relation to the broader atmospheric flow.

As the week progresses toward 11 February, the atmospheric structure becomes increasingly complex. Visualizing the atmosphere from the surface (1000 hPa) to the tropopause (200 hPa) reveals how high-altitude jet processes "pump" and strengthen the storm from above. Using Met.3D, we see that these surface lows are vertically coupled with upper-level flows that steer the systems toward land. This perspective allows us to locate the "ramps" of cold, dense air that force incoming warm, humid air to rise rapidly.

Identifying these frontal boundaries and moisture pathways is the critical starting point

for the rest of the group's research. These surface-level interactions create the thermodynamic environment needed to initiate the Warm Conveyor Belts (WCB) and Dry Intrusions (DI) that my teammates are investigating. By the 11th, as the system nears Europe, this moisture transport often concentrates into an "atmospheric river." Ultimately, this synthesis of surface and tropopause data provides the exact context needed to track the storm's evolution and identify the targets required for successful observational sampling.